

Telink

Zigbee Matter Dual-mode SDK Release Note

Telink Semiconductor

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Contents

Acknowledgements	2
Legal Disclaimer	2
Information	2
Revision History	2
1 Overview	3
2 Code Bases	3
2.1 Code Repository	3
2.2 Apply Patches	4
2.3 Change Device Type	5
2.4 Generate the Firmware	8
3 Matter Basic Functionality Test Results	9
4 Matter Ecosystem Test Results	10
4.1 Test Conditions	10
4.1.1 Telink Platform	10
4.1.2 Apple Ecosystem	14
4.1.3 Google Ecosystem	15
4.1.4 Samsung Ecosystem	16
4.1.5 Amazon Ecosystem	16
4.2 Matter Ecosystem Test	18
4.2.1 Matter Commissioning Test	18
4.2.2 Matter Interaction Test	18
4.2.3 Matter Stability Test	18
4.2.4 Matter Reconnection Test	19
4.2.5 Multi Fabric Test	19
5 Matter Certification Test Results	22
6 Appendix I	24

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Information

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Revision History

- **1.1.0:** Initial release.
- **1.1.1:** Fix heap overflow for amazon commissioning.
- **1.1.2:** Update release info and improve release note.

1 Overview

This document will introduce a zigbee matter dual-mode solution based on our TLSR952x(4MB) and TL321x(2MB) platform.

Matter v1.4 was officially released in March 2025, and the certification process for version 1.4 has also commenced. Telink has been closely monitoring the updates to Matter, actively participating in all TE, SVE, and Plugfest events organized by the CSA. Following the official release of Matter v1.4, we have been conducting comprehensive internal testing and decided on the code version to be made available for customer use.

In this document, we will provide detailed recommendations for Telink customers regarding the preferred code versions for development. Additionally, we will furnish specific results from internal testing, including certification tests and ecosystem tests.

Note: All the tests based on Telink TLSR952x(4MB) and TL321x(2MB) platform.

2 Code Bases

2.1 Code Repository

Our Zigbee Matter Dual-mode SDK includes the BLE/Zigbee Dual-mode SDK and Modified Matter SDK.

A complete Matter SDK consists of several repositories (Matter, Zephyr and so on). In this release, we recommend customers to use the following code bases for development on the Telink TLSR952x(4MB) and TL321x(2MB) platform:

- **Matter repository:** <https://github.com/project-chip/connectedhomeip/tree/master>
Commit ID: c0ee0a09b177bafd9255930de1043af1988bf4f7
Patch: zigbee_matter_dual_mode_sdk_matter_v1_1_2.patch
- **Zephyr repository:** https://github.com/telink-semi/zephyr/tree/develop_v3.3
Commit ID: 96b7ed873052518e90775af20b57805f69e9f195
Patch: zigbee_matter_dual_mode_sdk_zephyr_v1_1_2.patch
- **hal_telink repository:** https://github.com/telink-semi/hal_telink/tree/develop
Commit ID: 38c66795ef4a79453330a1aed85d1217843a8d05
Patch: zigbee_matter_dual_mode_sdk_hal_v1_1_2.patch

- **mcuboot repository:** <https://github.com/mcu-tools/mcuboot/tree/main>

Commit ID: 1558e7ab0aadb4eac11f03befb5ccd3fa3f0aafe

Patch: zigbee_matter_dual_mode_sdk_mcuboot_v1_1_2.patch

2.2 Apply Patches



Please follow the steps below strictly in the given order.

- Apply the Matter patch

1. Copy the `zigbee_matter_dual_mode_sdk_matter_v1_1_2.patch` to the path of `connectedhomeip` folder.
2. Apply this patch

```
1 cd ~/connectedhomeip
2 git apply zigbee_matter_dual_mode_sdk_matter_v1_1_2.patch
```

- Apply the zephyr patch

1. Copy the `zigbee_matter_dual_mode_sdk_zephyr_v1_1_2.patch` to the path of `zephyr` folder.
2. Apply this patch and update `hal`, `mcuboot`

```
1 cd ~/zephyrproject/zephyr
2 git apply zigbee_matter_dual_mode_sdk_zephyr_v1_1_2.patch
3 cd ..
4 west update
```

- Apply the `hal` patch and fetch `lib` file

1. Copy the `zigbee_matter_dual_mode_sdk_hal_v1_1_2.patch` to the path of `hal_telink` folder.
2. Apply this patch

```
1 cd ~/zephyrproject/modules/hal/telink
2 git apply zigbee_matter_dual_mode_sdk_hal_v1_1_2.patch
3 cd ../../..
4 west blobs fetch hal_telink
```

- Apply the `mcuboot` patch

1. Copy the `zigbee_matter_dual_mode_sdk_mcuboot_v1_1_2.patch` to the path of mcuboot folder.
2. Apply this patch

```
1 cd ~/zephyrproject/bootloader/mcuboot
2 git apply zigbee_matter_dual_mode_sdk_mcuboot_v1_1_2.patch
```

2.3 Change Device Type

By default, the lighting-app is a dimmable light, if you need to change the device type to extended color light, just need to change it via ZAP tool.

1. Run the ZAP tool

```
1 cd ~/connectedhomeip
2 source scripts/activate.sh
3 ./scripts/tools/zap/run_zaptool.sh examples/lighting-app/lighting-
  common/lighting-app.zap
```

2. Click Endpoint1 to see the device type

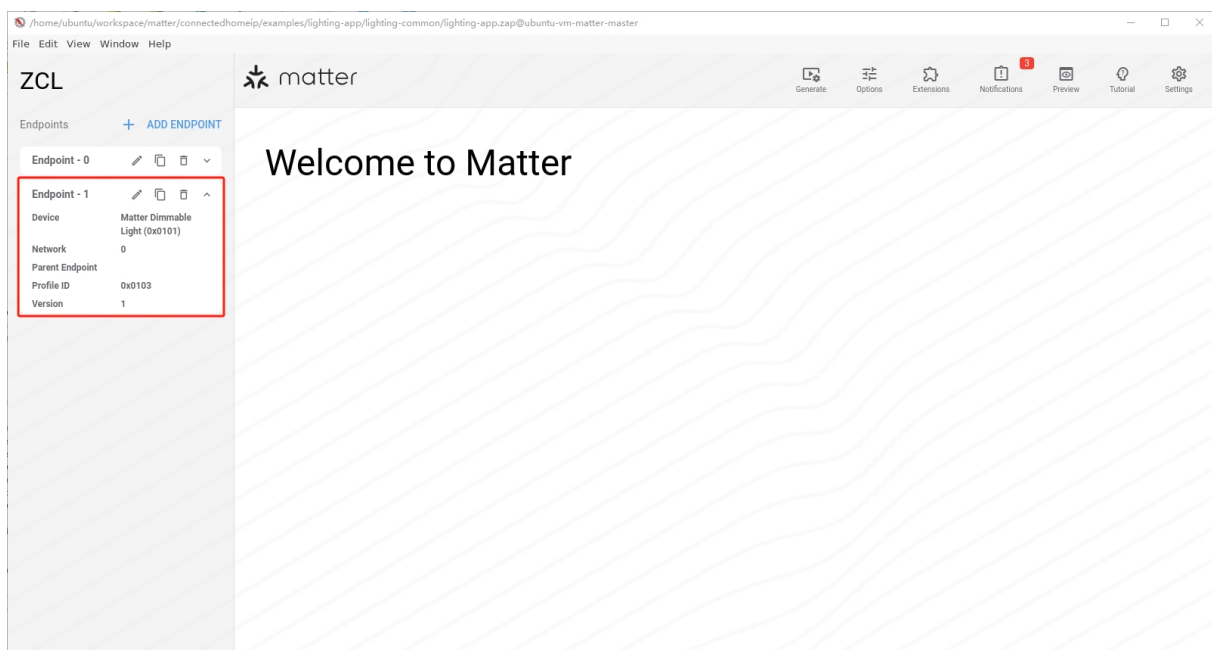


Figure 1: ZAP Endpoint1

2. Click **Edit** to change the device type in Endpoint1

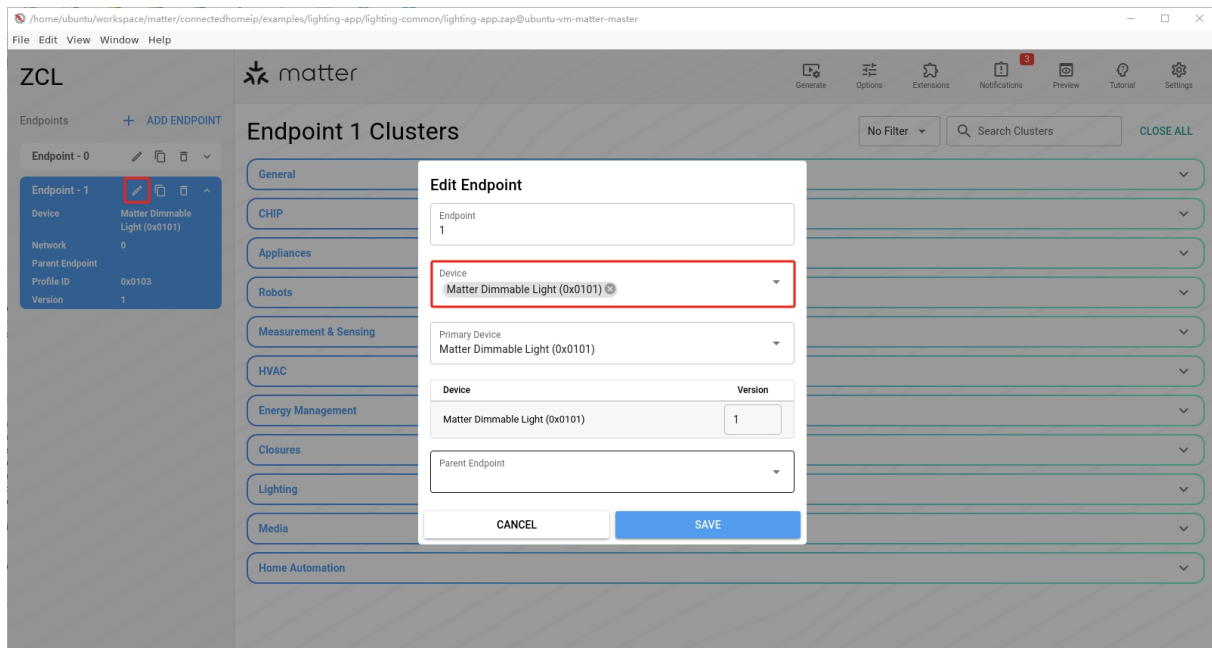


Figure 2: ZAP Change Device Type

3. Choose Matter Extended Color Light(0x10D)

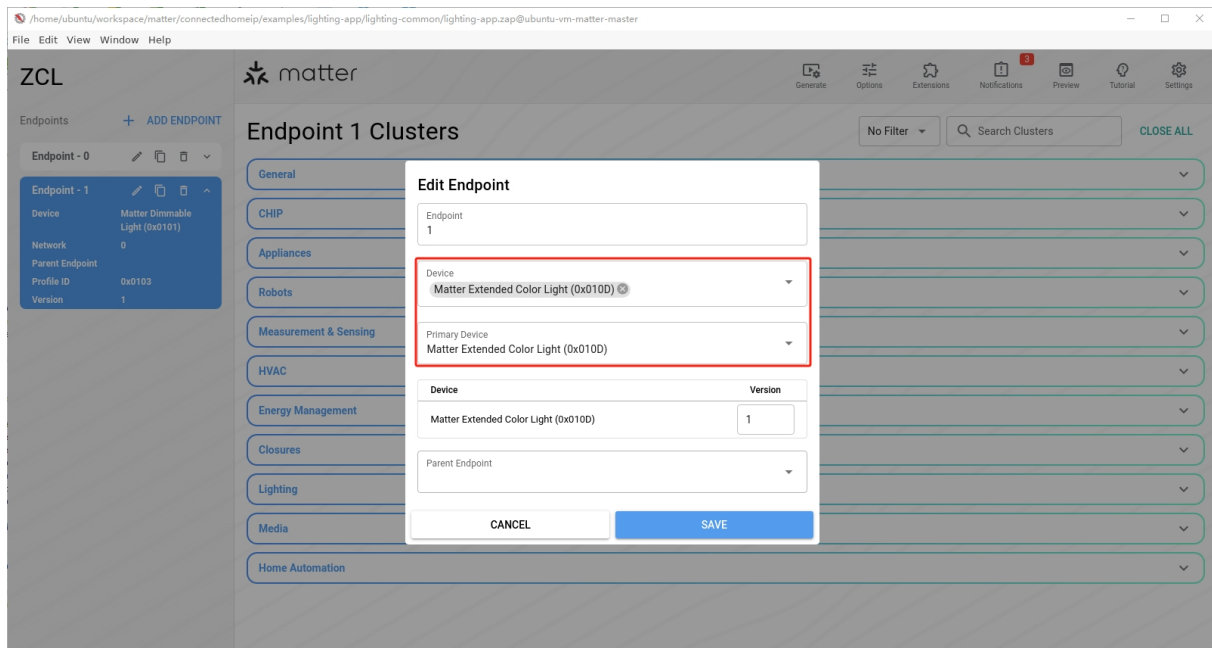


Figure 3: ZAP Choose Device Type

4. Save the changes

Click **File**–**save** in the top left corner of ZAP page or use **Ctrl+S** directly to save the changes.

2.4 Generate the Firmware

1. Compile command:

```
1 cd ~/connectedhomeip
2 source scripts/activate.sh
3 cd examples/lighting-app/telink
4 rm -rf build/
5 west build -p auto -b tlsr9528a -- -DFLASH_SIZE=4m
6 west build -p auto -b tl3218x
```

2. If you want to generate firmware that includes both Zigbee and Matter, please ensure the Zigbee firmware is present in the `zephyrproject/zephyr/ZB_FW/B92` and `zephyrproject/zephyr/ZB_FW/TL321X` folder and named `concurrent_sampleLight.bin`, and then execute the `west build` command.



You can check the detailed flash layout in [Appendix I](#).

If you want to get more information about how to set up a Matter environment, please check our [Telink Wiki page](#) and visit our developer's guide at the end of this webpage.

3 Matter Basic Functionality Test Results

We used the official TestHarness(version 2.11-beta1+fall2024) to conduct detailed tests on the basic functionality of the Matter device, including the following test items.

Test	Telink lighting-app
Factory Reset	PASS
Commissioning	PASS
OnOff Control	PASS
Reconnection	PASS
OTA Update	PASS

4 Matter Ecosystem Test Results

To simulate the everyday interactions users have with IoT devices, we conducted testing on specific device types within prominent commercial ecosystems such as Google, Apple, Samsung and Amazon.

Our simulated testing primarily includes the following tests:

1. **Commissioning Test** – Attempting commissioning operations using the same device types across the four ecosystems (Google, Apple, Samsung and Amazon).
2. **Interaction Test** – Controlling devices that have already been commissioned within each of the four ecosystems (Google, Apple, Samsung and Amazon).
3. **Stability Test** – Testing the duration for which devices that are already commissioned in each of the four ecosystems (Google, Apple, Samsung and Amazon) can maintain an online status.
4. **Reconnection Test** – Powering off devices that are already commissioned in each of the four ecosystems (Google, Apple, Samsung and Amazon), waiting for a certain period, and then powering them back on to test whether the devices can successfully reconnect.

4.1 Test Conditions

4.1.1 Telink Platform

We use Telink TLSR952x(4MB) and TL321x(2MB) EVB as the device under test, to test the behavior of different device types across four ecosystems.

**Figure 4:** TSLR952x-Development-Board-Front

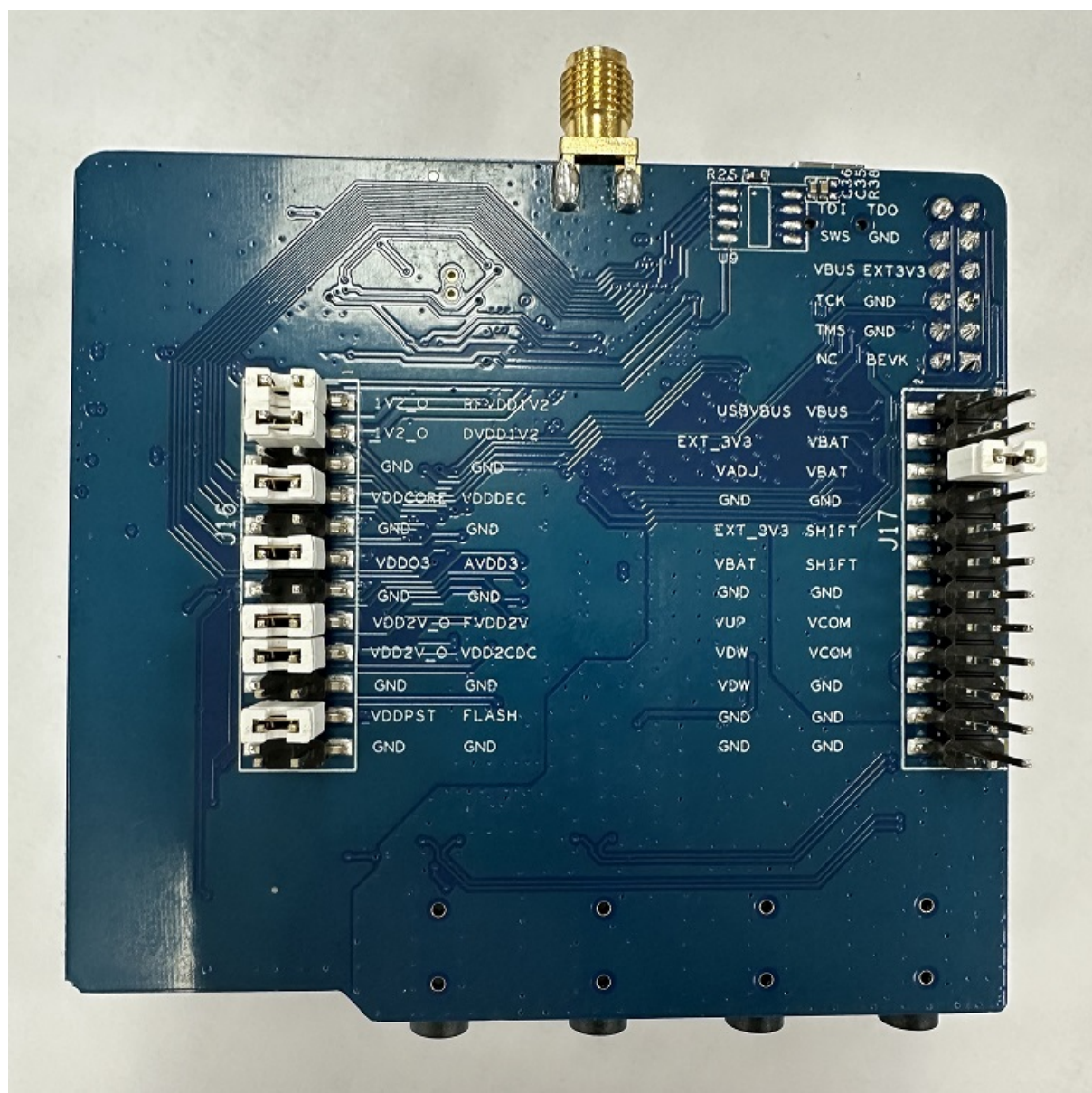
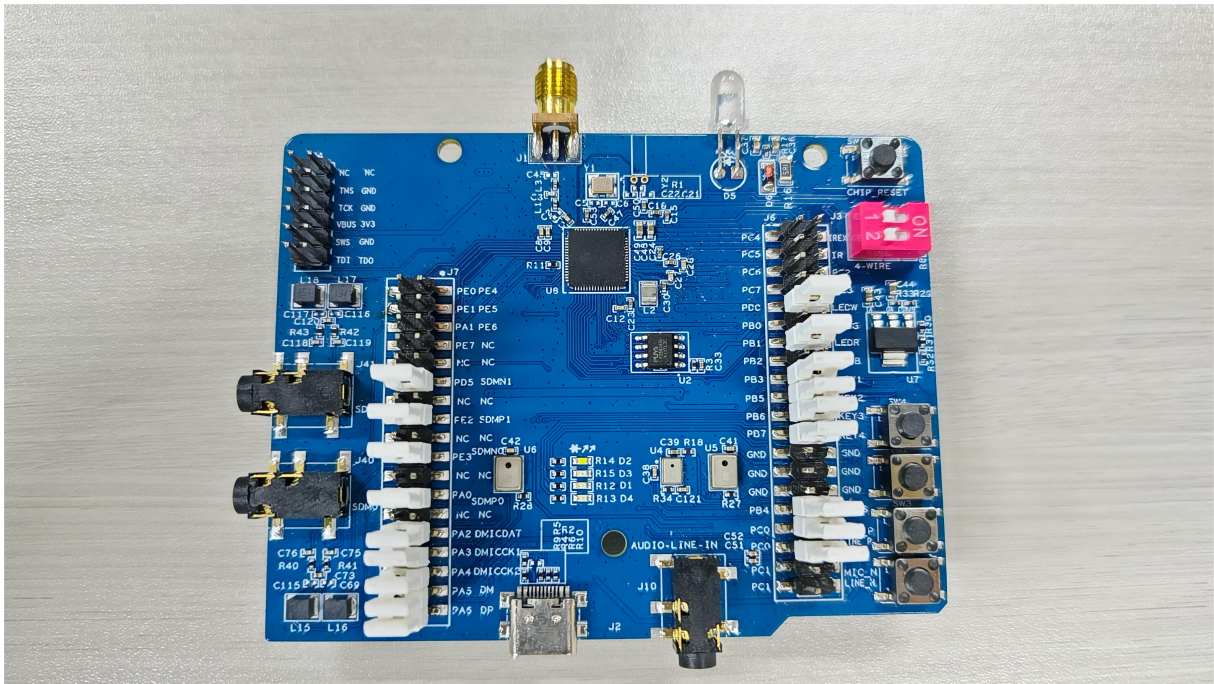


Figure 5: TLSR952x-Development-Board-Back

**Figure 6:** TL321x-Development-Board-Front**Figure 7:** TL321x-Development-Board-Back

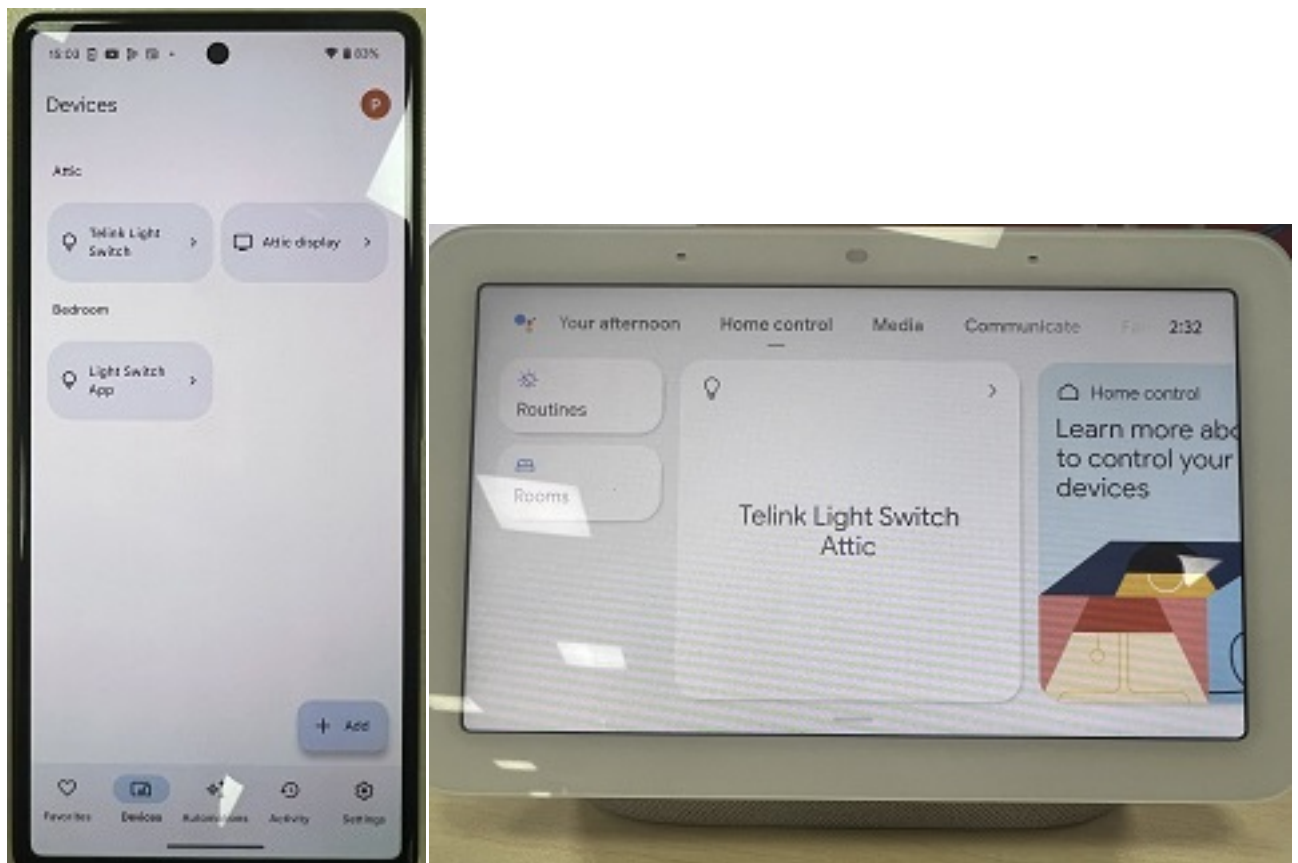
4.1.2 Apple Ecosystem

- iPhone 13
 - iOS version: iOS 17.5.1 (21F90)
- Apple Homepod mini
 - software version: 17.5 (21L569)



4.1.3 Google Ecosystem

- Pixel 6
 - Android version: 13
 - Google Play services: version 24.28.34 (190400-655627182)
- Google Nesthub (Gen 2)
 - Fuchsia Version: 18.20240225.3.63
 - Software Version: 58.102.2.613521407
 - Chromecast firmware version: 3.73.411796



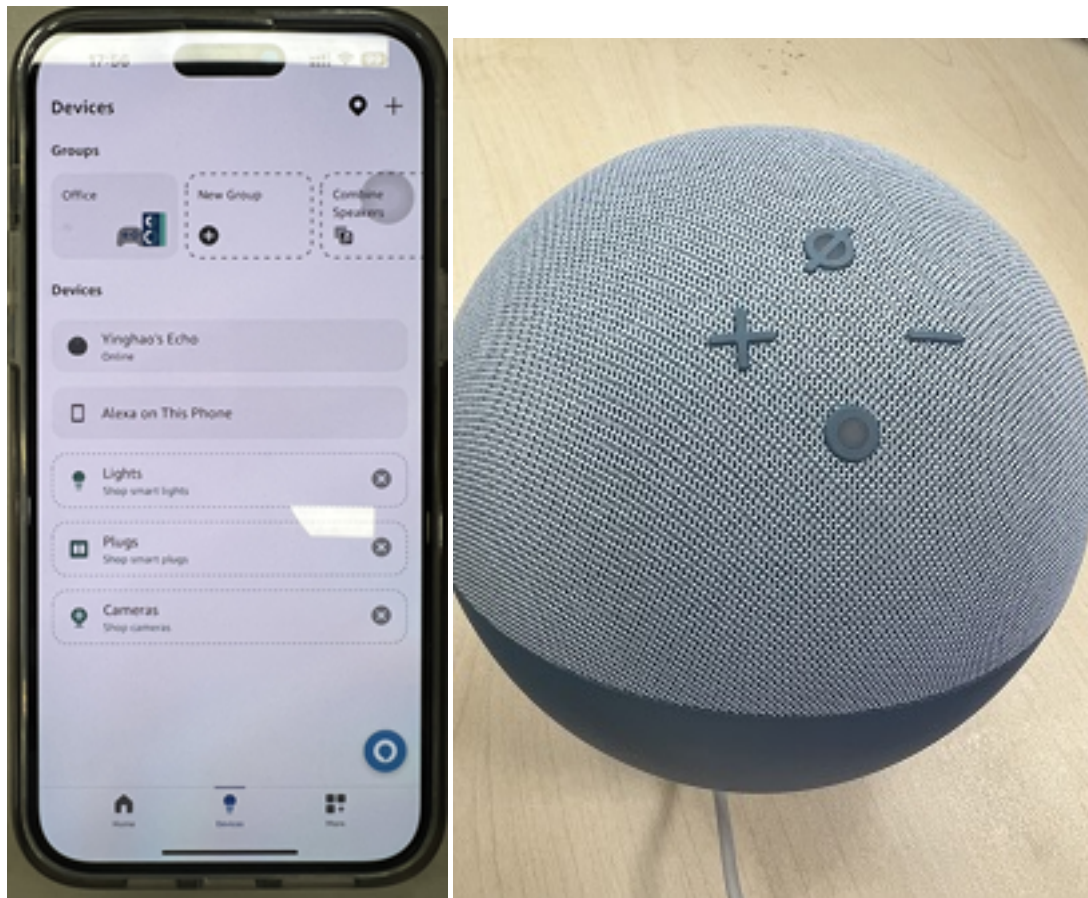
4.1.4 Samsung Ecosystem

- Pixel 6
 - Android version: 13
- Samsung SmartThings Station
 - Firmware version: P9501ZCU1AXF5



4.1.5 Amazon Ecosystem

- iPhone 14
 - iOS version: iOS 17.5.1 (21F90)
- Amazon Alexa Echo 4
 - Device Software Version: 0010369673092



4.2 Matter Ecosystem Test

4.2.1 Matter Commissioning Test

Test procedure:

1. Commission devices into different ecosystems.
2. Remove the commissioned devices.
3. Do factory reset.
4. Repeat Step 1 and 2 a total of 10 times continuously.

Test results:

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS	PASS

4.2.2 Matter Interaction Test

Test procedure:

1. Commission devices into different ecosystems.
2. Control the devices that has been commissioned into ecosystems. For example, with regards to a light, we would attempt to control its on/off using a mobile app within the respective ecosystems.

Test results:

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS	PASS

4.2.3 Matter Stability Test

Test procedure:

1. Commission devices into different ecosystems.

2. Leaving the commissioned devices in place for 7 days (7 * 24 hours) and periodically checking whether the devices can be reliably controlled during this period.

Test results:

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	On-Going	On-Going	On-Going	On-Going

4.2.4 Matter Reconnection Test

Test procedure:

1. Commission devices into different ecosystems.
2. Power off the devices, then power them on again. Afterward, check whether these devices can reconnect to the networks or not.
3. Repeat Step 1 and 2 a total of 10 times continuously.

Test results:

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS	PASS

4.2.5 Multi Fabric Test

1. Apple Ecosystem Multi Fabric

Test procedure:

- i. Commission the device into Apple HomePod-mini.
- ii. Share the device with Google, Samsung and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

Device Type	Google NestHub	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS

2. Google Ecosystem Multi Fabric

Test procedure:

- i. Commission the device into Google Nesthub.
- ii. Share the device with Apple, Samsung and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

Device Type	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS

3. Samsung Ecosystem Multi Fabric

Test procedure:

- i. Commission the device into Samsung SmartThings Station.
- ii. Share the device with Apple, Google and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

Device Type	Google NestHub	Apple HomePod-mini	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS

4. Amazon Ecosystem Multi Fabric

Test procedure:

- i. Commission the device into Amazon Alexa.
- ii. Share the device with Apple, Google and Samsung ecosystems, make sure the device can be controlled by all four ecosystems.

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station
lighting-app	PASS	PASS	PASS

5 Matter Certification Test Results

In order to bring Matter devices to market, it is imperative that they successfully undergo the Matter certification test, resulting in the issuance of a certificate by CSA. To achieve this, adherence to the latest Test Plans and test suites provided by the Matter community is mandatory.

To meet these requirements, we conducted a series of certification tests for the dual mode lighting app, selected a large number of automatic and python tests, and finally our device could pass all of these test cases. Here's the list of tested cases:



TestHarness version: 2.11-beta1+fall2024

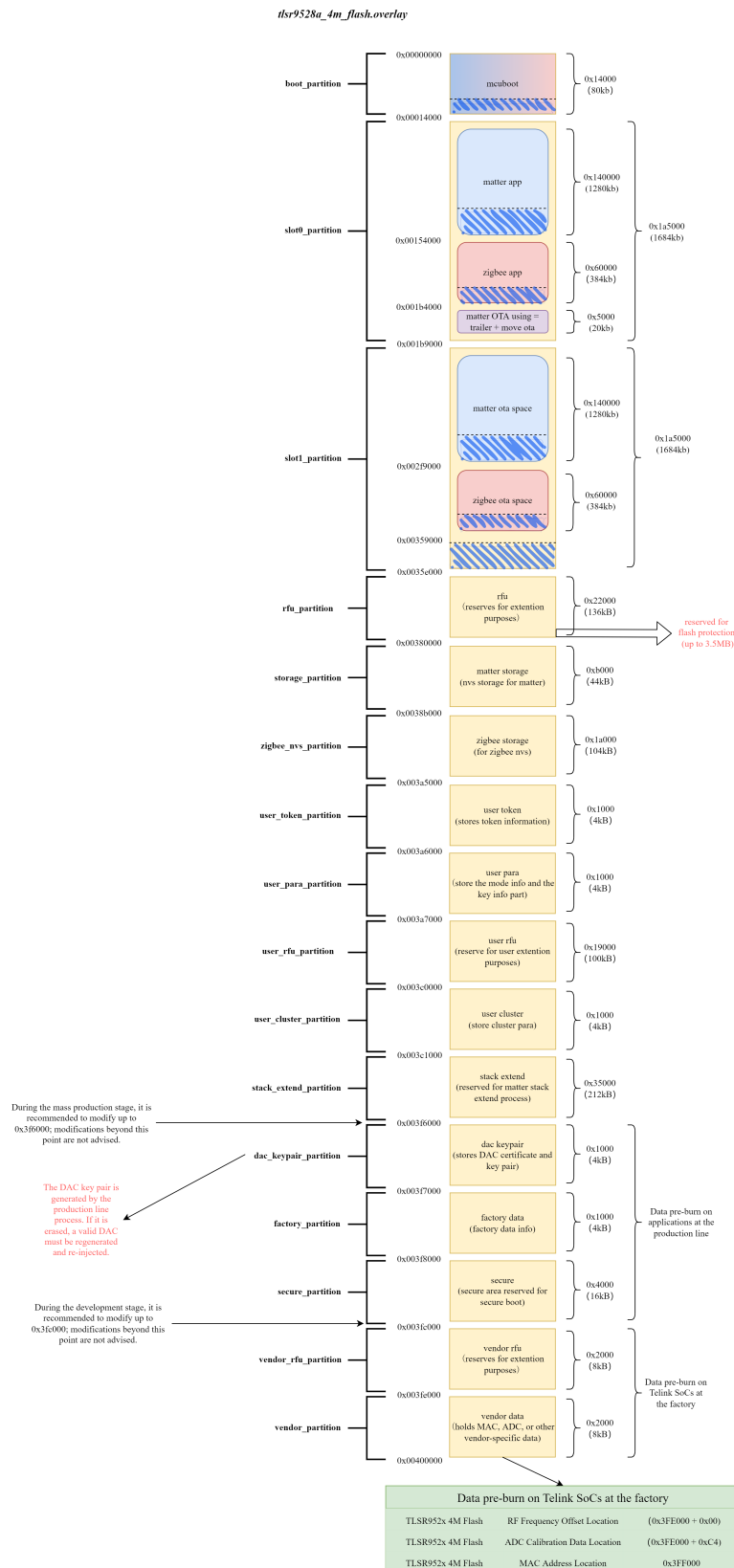
Test	Telink lighting-app
TC-ACE-1.1	PASS
TC-ACE-1.2	PASS
TC-ACE-1.3	PASS
TC-ACE-1.4	PASS
TC-ACE-1.5	PASS
TC-ACE-1.6	PASS
TC-ACE-2.1	PASS
TC-ACE-2.2	PASS
TC-ACL-2.1	PASS
TC-ACL-2.2	PASS
TC-ACL-2.3	PASS
TC-ACL-2.6	PASS
TC-CGEN-2.1	PASS
TC-CGEN-2.4	PASS
TC-CNET-1.3	PASS
TC-CNET-4.6	PASS
TC-DESC-2.2	PASS

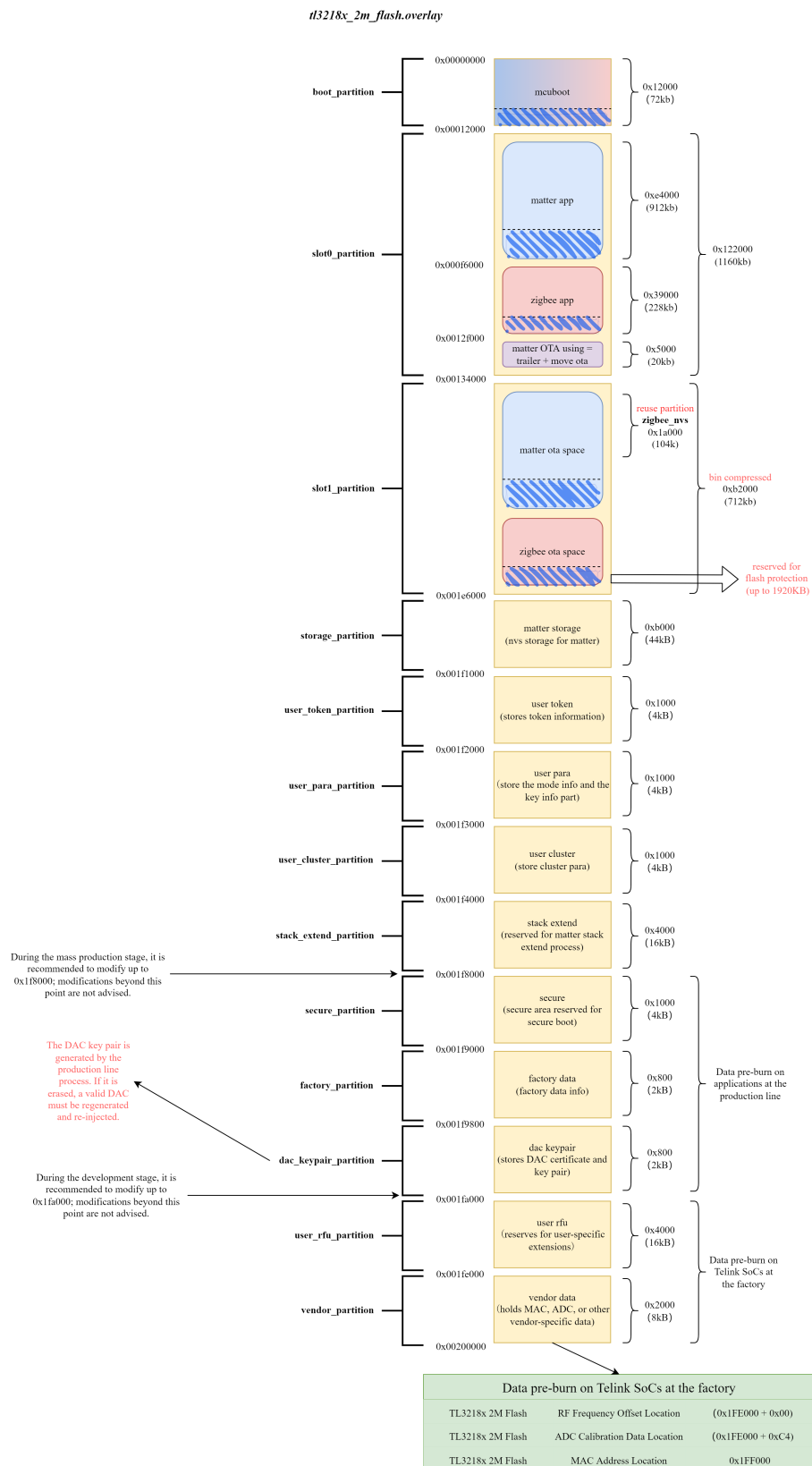
Test	Telink lighting-app
TC-DGGEN-2.3	PASS
TC-DGGEN-2.4	PASS
TC-DGGEN-3.1	PASS
TC-DGTHREAD-2.2	PASS
TC-DGTHREAD-2.3	PASS
TC-DT-1.1	PASS
TC-GRPKEY-2.1	PASS
TC-IDM-1.2	PASS
TC-IDM-1.4	PASS
TC-IDM-10.1	PASS
TC-IDM-10.3	PASS
TC-IDM-11.1	PASS
TC-LTIME-1.2	PASS
TC-OPCREDS-3.1	PASS
TC-SC-3.6	PASS
TC-SC-5.1	PASS
TC-SM-1.1	PASS
TC-SM-1.2	PASS
TC-ULABEL-2.1	PASS
TC-ULABEL-2.2	PASS
TC-ULABEL-2.3	PASS
TC-ULABEL-2.4	PASS
TC-G-2.1	PASS
TC-I-2.1	PASS
TC-OO-2.1	PASS
TC-OO-2.3	PASS
TC-OCC-2.1	PASS

Test	Telink lighting-app
TC-OCC-2.3	PASS
TC-RR-1.1	PASS
TC-CC-4.1	PASS
TC-CC-4.2	PASS
TC-CC-4.3	PASS
TC-CC-4.4	PASS
TC-CC-5.1	PASS
TC-CC-5.2	PASS
TC-CC-5.3	PASS
TC-CC-6.1	PASS
TC-CC-6.2	PASS
TC-CC-6.3	PASS
TC-CC-7.1	PASS
TC-CC-7.2	PASS
TC-CC-7.3	PASS
TC-CC-7.4	PASS
TC-CC-9.1	PASS
TC-CC-9.2	PASS
TC-CC-9.3	PASS

6 Appendix I

TLSR952x(4MB) and TL321x(2MB) Flash Layout:

**Figure 8:** TLSR952x Flash Layout

**Figure 9:** TL321x Flash Layout